

A manifest of docs/reports

Fifty-eight independent mathematical research packages,
classified by subject

Vladimir Reshetnikov

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1 Scope and conventions

This document catalogues the fifty-eight research packages stored under `docs/reports`. They arrived as sixty-four ZIP archives, which have been unpacked and grouped into the subject hierarchy described in Section 2. The archives themselves are no longer kept in the repository.

Sixty-four archives yield fifty-eight reports because six pairs turned out to prove the same theorem — in five of the six, by the same argument, down to shared lemmas and in one case shared worked counterexamples. Each pair has been merged into one report that proves the shared theorem once and keeps everything both originals built on top of it. Where the two reached a result by genuinely different routes, both proofs are kept and marked as alternatives, with a sentence on what each one buys. A merged entry below names both of its source archives and says what was doubled.

Every package is self-contained and follows the same shape: a typeset article (`*.pdf`) with its complete $\text{\LaTeX}$ source, a `README.md` summarising the result, usually a `code/` directory with exact-arithmetic verification scripts, a `data/` or `results/` directory holding the recorded output of those scripts, and often `STATUS.md`, `sources.md` or `notes/` files auditing the sources consulted and the precise limits of the claim.

Two cautions apply uniformly and are therefore not repeated in the individual entries. First, these are AI-assisted research drafts: none has been independently refereed, and none has been checked in a proof assistant. Second, the descriptions below record *what each report sets out to settle and what it claims to prove*. Compiling this manifest involved reading each package’s statement of its problem and result; it did not involve checking any proof. Where a report itself flags a limitation — a prior disproof, an imported theorem, a correction to the source, or an exception left open — that is noted, because it changes what the report is.

The naming of the directories is editorial. Several distinct reports attack the same published problem, and some arrived under colliding archive names, so each directory has been given a name describing its own angle on the problem; the merged directories are named for their combined scope, which is why none of them still carries the name that once distinguished it from its partner. The original archive names are recorded with every entry.

2 The directory tree

Category	Subcategory	Reports
ordinals-and-order-types	friendly-order-types	3
	wqo-powersets-and-statutes	4
	transfinite-words	2
	ordinal-arithmetic	2
	games-on-ordinals	2
hankel-determinants	somos-and-elliptic	3
	catalan-and-ballot	2
	growth-and-runs	2
	(top level)	1
congruences-and-valuations	supercongruences	2
	iterated-series	2
	(top level)	3
tetration-and-digit-stabilization	—	6
log-concavity-and-unimodality	—	7
graph-theory	—	3
enumerative-combinatorics	—	8
generating-functions-and-asymptotics	binary-product-coefficient-counts	3
	(top level)	3
Total		58

More than a quarter of the collection consists of counterexamples and corrections rather than proofs. Ordinal Chomp, open-query games, compositional tree series, the secant numbers, the decimal-length bound for tetration stabilization, power-tower stabilization, iterated-power digit stabilization, cluster variables, chromatic coefficients of cycles, products of q -integers, domination roots, Hardinian array constants, the mixed-base counterexample, recursive exponents, Gregory sign thresholds and binary substitution discrepancy all refute, or replace with a proved statement, an assertion published or catalogued elsewhere. Several reports whose main outcome is a proof still correct the statement they prove: the Somos-8 certificates, the shifted Catalan Hankel polynomials and the Rueppel determinants record a sign error, a missing one-unit shift and a numbering ambiguity in their sources, and the independence-system thresholds report refines its target after finding that the original conjecture had already been disproved by someone else.

3 Ordinals and order types

3.1 Friendly order types

Three reports on Isa Vialard’s *friendly order type* $f(P)$, the rank of the tree of open-ended bad sequences in which every chosen point has an incomparable partner in the current residual. All three prove the same finite formula; they differ in what they build on top of it, and in how they prove it — one by a forest-rank monovariant that survives moves deleting a whole upset, one through a lemma bounding the cut maxima. A fourth report proved the formula by the first report’s argument almost verbatim and has been merged into it.

1. Friendly Order Types Through Incomparability Components

ordinals-and-order-types/friendly-order-types/
incomparability-components-and-infinite-tails/friendly_order_types.pdf
from friendly_order_type_research (1).zip

The same component formula, together with an exact friendly rank for connected infinite well-partial-orders: writing $o(C) = \lambda + n$ with λ the limit part and $T(C)$ the finite set of points with finite principal upsets. Includes an exhaustive finite verification and a proof audit.

2. Friendly Order Type Through Incomparability Components

ordinals-and-order-types/friendly-order-types/
matroid-rank-and-ordinal-sums/article.pdf
from friendly_order_type_research (3).zip

Identifies $|P| - c(P)$ with the graphic-matroid rank and the incidence-matrix rank of the incomparability graph. Adds a substitution formula, a finite Cartesian-product formula, and additivity over arbitrary well-ordered ordinal sums; evaluates every well-partial-order whose incomparability components are finite, and gives countable counterexamples delimiting stronger infinite extrapolations.

3. Friendly Order Types of Finite Posets and Ordinal Products

ordinals-and-order-types/friendly-order-types/
components-forests-and-ordinal-products/friendly_order_types.pdf
from friendly_order_types_research.zip + friendly_order_type_research.zip

Proves $f(P) = |P| - c(\text{Inc}(P))$ for a finite poset P , where c counts connected components of the incomparability graph (isolated vertices included), and classifies the witnesses: every maximum friendly subset is P minus one minimal root per component, counted by $\prod |\text{Min}(C_i)|$. With Vialard's width-transfer theorem this gives ordinal width $\omega^{f(P)}$ for the Dershowitz–Manna multiset ordering. On the transfinite side it proves $f(\alpha_1 \times \cdots \times \alpha_r \times B) = \Theta$, the natural product, except that it is the predecessor of Θ when every α_i is a successor and B has a greatest element, and classifies arbitrary finite ordinal boxes. Also: spanning-forest certificates, exact monotonicity under added comparisons, a bivariate generating function, and explicit four-point factors with equal (o, h, w, f) whose products differ. *Merged; the Cartesian-product formula is proved twice, once by corner deletion and once by a chain-grid spanning-subgraph argument.*

3.2 Well-quasi-ordered powersets and statures

Four reports on the open questions in Section 6 of Abriola, Halfon, Lopez, Schmitz, Schnoebelen and Vialard, *Measuring well-quasi-ordered finitary powersets* (arXiv:2312.14587v2), and on the corresponding topological invariant.

4. Cofinal strata and ordinal absorption

ordinals-and-order-types/wqo-powersets-and-statures/
cofinal-strata-of-finitary-powersets/article.pdf
from cofinal_strata_ordinal_powersets.zip

For $S_\alpha(Q)$, a finite poset Q with each point replaced by a chain $\alpha = \omega^\rho$, proves $o(K(S_\alpha(Q))) = \sum_k \omega^{\rho \otimes k} c_k(Q)$ and $h(K(S_\alpha(Q))) = \alpha \cdot \text{height}(M(Q))$, where c_k counts size- k maximal antichains not dominated by a larger one. The maximal-order-type theorem allows nonuniform pure well-partial-order fibers and is proved by a finite cofinal-stratification argument.

5. Intrinsic ordinal approximations for finitary powersets

ordinals-and-order-types/wqo-powersets-and-statures/
intrinsic-ordinal-approximations/article.pdf
from intrinsic_ordinal_approximations.zip

Studies the intrinsic invariant $a_o(A) = \sup\{o(B) + 1 : \text{Idl}(B) \hookrightarrow A\}$. Proves the universal bound $a_o(A) \leq h(P_f(A))$ with equality exactly when every finitely generated downset of A is an algebraic depo, gives the closed formula $h(P_f(\prod_i \alpha_i)) = \sup(\otimes_i (1 + a_i) + 1)$, and shows the universal identity fails on $\lambda + F$.

6. Ordinal Order Maps and Finitary Powersets of Ordinal–Finite Grids

[ordinals-and-order-types/wqo-powersets-and-statures/
ordinal-order-maps-and-grids/ordinal_order_maps.pdf](#)
from [ordinal_order_maps_research.zip](#)

Exact maximal order types and heights for finite Hoare powersets of $\beta \times Q$ with Q a finite poset: for $\beta = \omega^{\gamma_0} + \dots + \omega^{\gamma_{m-1}}$, $o(\text{Mon}(Q, \beta)) = \bigoplus_{s \text{ isotone}} \omega^{\bigoplus_q \gamma_{s(q)}}$.

7. Sharp ordinal bounds for Hoare powerspaces

[ordinals-and-order-types/wqo-powersets-and-statures/
hoare-powerspace-statures/article.pdf](#)
from [hoare_statures_research.zip](#)

For all Noetherian spaces, $1 + \|X\| \leq \|HX\| \leq 2^{\|X\|}$, with witnesses attaining both endpoints at every ordinal and T_1 upper witnesses; the older ω^α bound is attainable at infinite input stature α exactly when $\omega \cdot \alpha = \alpha$; the complete spectrum at $\omega + n$ is $\omega \cdot j(U) + (j(Q) - j(U))$. Extends the known well-quasi-order results of Abriola et al. to arbitrary Noetherian spaces.

3.3 Transfinite words

Two reports on Harry Altman, *Bounding finite-image sequences of length ω^k* (arXiv:2409.03199v2), Example 3.15: is the bound ω^{ω^4} sharp for words of length $< \omega^2$ over the two-element antichain? A third report matched the first lemma for lemma and has been merged into it.

8. Exact maximal order types of finite-alphabet transfinite words

[ordinals-and-order-types/transfinite-words/
finite-alphabet-transfinite-words/article.pdf](#)
from [finite_alphabet_transfinite_words.zip](#)

Proves $o(W_k(P)) = \omega^{\omega^{N_k(P)-1}}$ for words of ordinal length $< \omega^k$ over a finite poset P under subsequence embedding, where N_k counts indecomposable classes of a canonical finite atom poset $I_k(P)$, with $N_1(P) = |P|$ and $N_{k+1}(P) = |P| + J(I_k(P)) - 1$. For $k = 2$ the binary antichain gives ω^{ω^4} — the tightness Altman leaves open — and the binary chain gives ω^{ω^3} .

9. Guarded periodic blocks and exact maximal order types below omega squared

[ordinals-and-order-types/transfinite-words/order-types-below-omega-squared/article.pdf](#)
from [omega2_order_types_proposed_solution.zip](#) + [ordinal_words_research.zip](#)

Proves $o(s_{\omega^2}(P)) = \omega^{\omega^{n+j(P)-2}}$ for $n = |P| \geq 2$, with $j(P)$ the number of downsets including the empty one; the empty and singleton alphabets are genuine exceptions. The binary antichain gives ω^{ω^4} — the tightness Altman leaves open — and the binary chain ω^{ω^3} . Adds an extremality corollary with both equality characterisations (among n -element posets only the chain minimises and only the antichain maximises), the $j(P)$ toolkit, the arbitrary-support-family classification, and an exact-length stratum calculus showing the supremum over lengths is only ω^{ω^n} , strictly below the answer — so the theorem cannot be obtained stratum by stratum. *Merged; the product amplification is done twice, once via de Jongh–Parikh and once by an elementary hand proof that removes the import.*

3.4 Ordinal arithmetic

Two independent proposed solutions to Problem 6.2 of Paolo Lipparini, *Monotone infinitary operations on ordinals* (arXiv:2505.00424v2): define and study the least weakly monotone, e -special strictly increasing operation on countable sequences of ordinals.

10. An Explicit Formula for the Minimal e -Monotone Infinitary Ordinal Sum

[ordinals-and-order-types/ordinal-arithmetic/lipparini-minimal-infinitary-sum/article/
explicit_ordinal_sum.pdf](#)
from [lipparini_problem_6_2_proposed_solution.zip](#)

An explicit formula for the least admissible operation, proved by an arithmetic admissibility upper bound and a rank-theoretic minimality lower bound, using an explicit countable-multiset

profile order, its rank, absorption and correction rules, bounded-poset heights and finite-cap ranks. Imports Lipparini’s formula and minimality theorem for the weaker operation S .

11. An Explicit Formula for Lipparini’s Minimal Infinitary Ordinal Operation

[ordinals-and-order-types/ordinal-arithmetic/
lipparini-minimal-operation-formula/article.pdf](#)
from [ordinal_operation_research.zip](#)

A three-case finite ordinal-arithmetic formula with proofs of admissibility and pointwise minimality. Relative to Lipparini’s already evaluated S , the correction is zero, a finite increment, or one *natural* summand ω — ordinary addition of ω is not a substitute.

3.5 Games on ordinals

12. A counterexample to winner stability in ordinal Chomp

[ordinals-and-order-types/games-on-ordinals/
ordinal-chomp-winner-stability/ordinal_chomp_counterexample.pdf](#)
from [ordinal_chomp_counterexample.zip](#)

For the numerical semigroup $S = \langle 4, 6, 9 \rangle$, Chomp on S is a second-player win while Chomp on the ordinal square $S^2 = \{\omega a + b : a, b \in S\}$ is a first-player win — the opening $\omega \cdot 4 + 4$ leaves Grundy value zero, and the same move wins at every exponent ≥ 2 , so $\text{ch}(4, 6, 9) = 2$. This refutes Conjecture 4.1 of arXiv:2504.07317v1 and answers Question 4.6 of v3 negatively, already for natural-number generators.

13. Open-query games on ordinal spaces

[ordinals-and-order-types/games-on-ordinals/open-query-membership-games/article.pdf](#)
from [ordinal_membership_games.zip](#)

A negative answer to Problem 1.3 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3): two copies of $[0, \omega]$ each have set-membership number 1, but their topological sum has value 2. The general theorem computes the invariant for every nonempty Hausdorff space of finite Cantor–Bendixson height h as $\lceil \log_2(h + 1) \rceil$ when the last nonempty derivative is a singleton and $\lceil \log_2(h + 2) \rceil$ otherwise.

4 Hankel determinants and continued fractions

4.1 Somos sequences and elliptic curves

14. A proof of Barry’s Hankel–Somos-6 conjecture

[hankel-determinants/somos-and-elliptic/barry-somos-6-hankel-recurrence/article.pdf](#)
from [barry_somos6_research.zip](#)

Conjecture 7 of Paul Barry, arXiv:2211.12637: for the Hankel determinants H_N of a four-parameter Jacobi-type generating function, $H_N H_{N-6} = \alpha H_{N-1} H_{N-5} + \gamma H_{N-3}^2$ with $\alpha = t^2(r + 2)^2$ and $\gamma = t^3((s + t)(r + s + t + 2)^2 + t(r + 2)^3)$. Also proves a homogeneous four-parameter form and a seven-parameter polynomial Hankel theorem making exceptional specialisations rigorous; the verifier never divides by a determinant.

15. Finite Hankel Certificates for Three Somos-8 Conjectures

[hankel-determinants/somos-and-elliptic/somos-8-hankel-certificates/article.pdf](#)
from [somos_hankel_proofs.zip](#)

Conjectures 11, 12 and 13 of the same Barry paper, by importing Hone’s genus-two Hankel theorem (arXiv:1907.05204v3, Thm. 5.4) and supplying a binomial translation, a polynomial-minor specialisation valid on the degenerate locus, and a rank-bound lemma that turns four symbolic rows into an all-index identity. Identifies a sign error in the source’s Catalan expression for Conjecture 13.

16. Elliptic addition and Jacobi continued fractions

hankel-determinants/somos-and-elliptic/
elliptic-addition-and-jacobi-fractions/article.pdf
from elliptic_continued_fraction_research.zip

A consistently indexed, non-torsion version of Barry's 2023 Conjecture 4, with the coordinate formulas of Conjecture 3: every continued-fraction tail is given explicitly and removing one layer is shown to be elliptic translation. Hypotheses: characteristic zero, $Y^2 + aXY + bY = X^3 + cX^2 + dX$ nonsingular, $b \neq 0$, and $P = (0, 0)$ of infinite order; order-three and order-four examples show the regular fraction genuinely needs the last one.

4.2 Catalan and ballot moments

17. Shifted Catalan Hankel Polynomials

hankel-determinants/catalan-and-ballot/shifted-catalan-hankel-polynomials/article.pdf
from catalan_hankel_conjecture_solution.zip

For $D_N^{(m)}(a, b) = \det(aC_{i+j+m} + bC_{i+j+m+1})$, proves the denominator-exponent assertion on page 4 of Barry, arXiv:2011.10827v1, and shows the separately printed Conjecture 2 needs an explicit one-unit shift correction. The minimal characteristic polynomial is $(X^2 - (a+2b)X + b^2)^{m(m-1)/2+1}$, so the minimal order is $m(m-1) + 2$; all exceptional reduced denominators, the exact coefficient formula, numerator degree and reciprocal symmetry are supplied.

18. Product formulas for ballot-polynomial Hankel determinants

hankel-determinants/catalan-and-ballot/
ballot-polynomial-hankel-determinants/article.pdf
from Cigler_Hankel_Conjectures_13_15.zip

Conjectures 13, 14 and 15 in Section 5 of Johann Cigler, arXiv:2111.14492v3. An explicit coefficient product gives the sign and exact support of the normalised shift- k Hankel determinant of the ballot moments and positivity of every residual coefficient for $n \geq 1$; the same products give the conjectured generating-function denominators, exact numerator degrees and reciprocal numerator identities. Adds strict log-concavity of the residual coefficients and their limiting binomial profile.

4.3 Arithmetic Hankel determinants

The numerator formula recorded as a conjecture in OEIS A122251, with A122249 as its binary special case. Two reports attacked it and proved it the same way, sharing even their worked counterexamples; they are merged here into one. The merged report sits at the category's top level.

19. A Proof of an OEIS Hankel-Numerator Conjecture

hankel-determinants/a122251-numerators-and-denominators/report.pdf
from hankel_numerator_research.zip + hankel_oeis_conjecture_proof.zip

For $H_n(m, a) = \det(1/(a+m(i+j)))_{0 \leq i, j \leq n}$, the reduced numerator is $\prod_{p|m} p^{n(n+1)v_p(m)+2 \sum_{k \leq n} v_p(k!)}$: independent of the coprime offset a , a perfect square, and supported on the primes dividing m . Taking $m = p$ and $a = 1$ settles the OEIS A122251 conjecture, with A122249 as the binary case. Also gives a closed form for the reduced denominator and all its local valuations, a sharp gcd theorem over the allowed offsets realised at just two explicitly constructed shifts, the extension to unequal row and column steps $\det(1/(a + m_1i + m_2j))$, recurrences, digit-sum asymptotics, and a Lambert-series generating function with a natural boundary. *Merged; the determinant is evaluated three independent ways, and a second proof of the numerator formula runs through an explicit inverse matrix and MacMahon's boxed plane partitions.*

4.4 Growth laws and run statistics

20. A proof of the Apéry Hankel growth conjecture

[hankel-determinants/growth-and-runs/](#)

[apery-hankel-determinant-growth/apery_hankel_growth.pdf](#)

from [apery_hankel_research.zip](#)

For $A_k = \sum_j \binom{k}{j}^2 \binom{k+j}{j}^2$ and $D_n = \det(A_{i+j})$, proves $\log D_n = n(n+1) \log \frac{17+12\sqrt{2}}{4} + O(n)$ — the limit conjectured in the September 2026 revision of OEIS A228143 — using Edgar’s moment-density theorem. Also an explicit law for $\det(A_{m(i+j)+r_n})$ with $r_n/n \rightarrow \rho$, and non-D-finiteness of the formal generating series of D_n .

21. Binary Runs in Rueppel Hankel Determinants

[hankel-determinants/growth-and-runs/rueppel-binary-run-determinants/rueppel_hankel.pdf](#)

from [rueppel_hankel_research.zip](#)

For the Rueppel series $r(x) = \sum_{j \geq 0} x^{2^j-1}$, the polynomial identity $H_n(tx + 1/r(x^2)) = (-1)^{1+\varepsilon+\lfloor R(n)/2 \rfloor} (\varepsilon + (R(n) - \varepsilon)t^2)$, where $R(n)$ counts runs of equal bits in the binary expansion of n and ε is the low bit of the Gray code $n \oplus \lfloor n/2 \rfloor$. Settles statements 8, 10, 13 and 23 of Barry’s 2021 paper (not the identically numbered ones elsewhere), with a correction.

5 Congruences and p -adic valuations

5.1 Supercongruences

22. Congruences for OEIS A348410 at every prime

[congruences-and-valuations/supercongruences/a348410-all-prime-congruences/article.pdf](#)

from [a348410_supercongruence.zip](#) + [A348410_Supercongruence_Research.zip](#)

For $A_{\alpha,\beta}(n) = [x^n](1-x)^{-\alpha n}(1+x)^{-\beta n}$ proves the odd-prime cubic supercongruence $v_p(A(Np) - A(N)) \geq 3v_p(N) - \varepsilon_p$ with $\varepsilon_3 = 1$, $\varepsilon_p = 0$ for $p \geq 5$, and no coprimality condition; $(\alpha, \beta) = (2, 1)$ is Bala’s A348410 conjecture. Separately proves an exact binary leading-defect formula, $A(m2^r) - A(m2^{r-1}) \equiv 2^{3r-3} \alpha \beta \binom{\alpha+\beta+1}{m-1} \pmod{2^{3r-2}}$, hence the all-prime bound $3r - v_p(12)$ and sharp denominators 12 and 24. An appendix audits arXiv:2104.10754v1, whose framing conclusion and weighted harmonic assertion fail on its own displayed examples. *Merged; the odd-prime and binary halves run on different mechanisms — a doubling automorphism, and reflection with formal integration by parts — and both are kept.*

23. Signed binomial supercongruences and OEIS A352373

[congruences-and-valuations/supercongruences/](#)

[a352373-signed-binomial-supercongruences/signed_binomial_supercongruence.pdf](#)

from [OEIS_A352373_supercongruence.zip](#)

For $a(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$ proves $a(mp^h) \equiv a(mp^{h-1}) \pmod{p^{3h}}$ for $p \geq 5$ and $a(m3^h) \equiv a(m3^{h-1}) \pmod{3^{3h-1}}$, with m not required coprime to p . Covers A352373, A348410, A351856 and A351857 as specialisations; the only non-elementary import is the classical Jacobsthal binomial congruence.

5.2 Congruences for iterated and compositional series

24. Arithmetic of a Self-Iterating Generating Function

[congruences-and-valuations/iterated-series/a396807-self-iterating-series/article.pdf](#)

from [A396807_congruences_and_prime_power_lifts.zip](#)

Both conjectures in OEIS A396807 for $A(x) = x + A^{[5]}(x)A^{[6]}(x)$ (functional iteration): $A(x) \equiv x/(1-x)$ and $A^{[k]}(x) \equiv x/(1-kx)$ modulo 10 for every integer k , negative iterates included. Adds a sharp modulus theorem for $F = x + cF^{[r]}F^{[s]}$, rational lifts at every selected prime power, and minimal period 46860 for A396807 modulo 100 starting at index 1.

25. Prime-power congruences for compositional tree series

[congruences-and-valuations/iterated-series/](#)

[compositional-tree-series-congruences/article.pdf](#)

from `compositional_tree_congruences.zip`

For $A_l(x) = x \exp(A_l^{ol})$, shows the OEIS A396805 claim $a_5(n) \equiv n \pmod{5}$ is *false*: first failure $n = 23$, with exception set $\{20j + 3, 20j + 4 : j \geq 1\}$ and residue 2 throughout. Proves eleven other conjectural assertions in A396803–A396806 via a general reduction of coefficients mod p^α to typed rooted trees on fewer than p^α labels, yielding eventual periods, rational residue generating functions, and an exact classification of $a_l(n) \equiv n \pmod{p}$.

5.3 Valuations, divisibility and periodicity

26. Binary Carries and the Divisibility of Two-Stack-Sortable Permutation Numbers

`congruences-and-valuations/a000139-binary-carries-parity/research.pdf`

from `a000139_carries_research.zip`

Proves Peter Bala’s parity conjecture for OEIS A000139, $a(n) = 2(3n)! / ((n+1)!(2n+1)!)$: $a(n)$ is odd exactly when n is odd and its binary expansion contains no 11. The full valuation is $v_2(a(n)) = s(n) + s(n+1) - s(3n)$ in binary digit sums. Two independent parity proofs, plus exact valuation distributions, recurrences, asymptotics, extremal formulas and limit laws.

27. Polynomial divisibility of height-one factorial ratios

`congruences-and-valuations/factorial-ratio-polynomial-divisibility/article.pdf`

from `bala_factorial_divisibility_research.zip`

A general rational-root criterion for uniform polynomial divisibility of balanced integral factorial ratios, applied to Peter Bala’s uniform-product conjectures for OEIS A211417 and A295431: explicit multipliers for every product length, with the least multipliers certified in eight particular examples. Includes an atlas of the 52 sporadic height-one ratios with recurrences, generating functions and asymptotic expansions.

28. When the Secant Numbers Become Periodic

`congruences-and-valuations/secant-number-periodicity/article.pdf`

from `secant_periodicity_research.zip`

Refutes the OEIS A000364 conjecture that the secant numbers from $n = 1$ are purely periodic modulo every m : at $m = 27$, $a(1) \equiv 1$ but $a(19) \equiv 10$. The exact replacement is that pure periodicity from index one holds if and only if no odd prime cube divides m ; the least eventual period always divides $\varphi(m)$. Determines the least eventual period and exact onset for every modulus, with densities and a certified mean onset.

6 Tetration and digit stabilization

29. A Counterexample to a Decimal-Length Bound for Tetration Stabilization

`tetration-and-digit-stabilization/congruence-speed-decimal-length-bound/article.pdf`

from `tetration_counterexample.zip`

An explicit 2544-digit base $A = 1 + qu$ ($q = 5^{2547}$) for which $\nu_{10}(T_{n+1} - T_n) = \min(1 + 2548n, 2547(n+1))$ at every height, so the congruence speed at height $\text{len}(A) + 2$ is 2548 while the permanent value is 2547. This contradicts the decimal-length form of Conjecture 1 of Marco Ripà, *The congruence speed formula* (NNTDM 27(4), 2021). The report audits the later Ripà–Onnis confirmation: their weaker valuation bound is compatible with the counterexample, the decimal-length bound is not.

30. Tetration, perfect powers, and frozen decimal digits

`tetration-and-digit-stabilization/perfect-power-minima-for-frozen-digits/article.pdf`

from `tetration_perfect_power_minima.zip`

Determines $\rho(n)$, the smallest n th perfect power $x^n > 1$ whose eventual number of newly frozen terminal decimal digits is exactly n : $\rho(1) = 2$, $\rho(2) = 49$, and $\rho(n) = B(s)^n$ for $n \geq 3$ with

$s = n - v_2(n)$ and $B(s) \in \{2^s \pm 1, 3 \cdot 2^s \pm 1\}$ according to $s \bmod 4$. All other roots are ruled out by a deterministic quadratic size barrier plus small exact certificates.

31. Phase-shift classification for integer tetration

tetration-and-digit-stabilization/
phase-shift-classification/tetration_phase_classification.pdf
from tetration_phase_classification.zip

Proves the 21-word phase-shift conjecture in OEIS A376842, the 14-word even-base restriction from the appendix of Ripà's 2025 paper, and the 28-word prescribed-anchor conjecture in OEIS A376446. The mechanism compares the 2-adic and 5-adic valuations of $D_b = T_{b+1} - T_b$ at the first permanently constant-speed height: strict 2-adic dominance gives phase 5, strict 5-adic dominance a multiplicative orbit in $\mathbb{F}_5^\times$, and balanced valuations exactly four further anchored words.

32. A Sharp Convergence Theorem for the q -Newton Series of Subcritical Tetration

tetration-and-digit-stabilization/q-newton-series-convergence/article/
tetration_qnewton.pdf
from tetration_qnewton_research.zip

Proves convergence of the explicit q -binomial interpolation series of MathOverflow question 259278 for every real base $1 < a < e^{1/e}$, with $\log a = qe^{-q}$: the grouped series converges absolutely and locally uniformly for $\operatorname{Re} z > -2$, converges conditionally on $\operatorname{Re} z = -2$ except at $z = -2 + 2\pi ik/(-\log q)$, and diverges for $\operatorname{Re} z < -2$. Adds strict complete monotonicity and uniqueness under an eventual log-convexity condition.

33. Stabilization of integer power towers

tetration-and-digit-stabilization/power-tower-stabilization/article.pdf
from tetration_stabilization_research.zip + tetration_exact_digits.zip

Refutes the claim that every prime base ending in 9 reaches its permanent decimal congruence speed by height 2 (Ripà, Conjecture 2; arXiv:2208.02622v1, Conjecture 4.1): 2749 is the smallest prime counterexample, with $H(2749) = 3$. Proves exact formulas for $C_a(h)$, $V(a)$ and $H(a)$, an exact natural density for onsets, and infinitely many primes at each prescribed onset via CRT and Dirichlet. Also resolves the three conjectural comments in OEIS A324017 — two proved, the third disproved, with $3 \uparrow\uparrow 2 \equiv 27$ and $3 \uparrow\uparrow 3 \equiv 59 \pmod{64}$ — via $T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}$ for even $q \geq 4$, so exactly one radix- q digit freezes per height. Adds a factorization-free digit-lifting algorithm, modular saturation across Knuth arrow ranks, and a sign-sensitive local p -adic Lambert W formula. *Merged; the shared one-digit-per-height theorem was proved identically in both originals and is proved once here.*

34. Delayed digit stabilization: a counterexample and sharp bounds

tetration-and-digit-stabilization/
iterated-power-digit-stabilization/delayed_digit_stabilization.pdf
from digit_stabilization_research.zip

For $S_a(b) = v_{10}(a^{10^{b+1}} - a^{10^b})$ and $D_a(b) = S_a(b) - S_a(b-1)$, answers negatively the question whether $D_a(b) = 1$ for all $b \geq 3$: the smallest counterexample is $a = 3 \cdot 2^{99} + 1$, with $S_a(2) = 100$, $S_a(3) = 102$, $D_a(3) = 2$. Proves minimality, an exact onset formula, the smallest counterexample at every stage, the natural density of exceptions, and arbitrary-radix extensions. The article is explicit that a^{10^b} is not genuine tetration and derives the tower formulas separately.

7 Log-concavity, unimodality and positivity

35. Single curves do not force log-concavity

log-concavity-and-unimodality/cluster-variable-log-concavity-counterexample/article.pdf
from annular_single_lamination_counterexample.zip

An unpunctured annulus with one marked point on each boundary and a single unit-weight simple lamination curve gives, after mutations 2, 1, 2 with the two mutable variables specialised to 1, the

polynomial $1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6$; the consecutive coefficients 3, 2, 4 form a strict valley, refuting Conjecture 1 of arXiv:2508.04396v3 and conflicting with its general unpunctured-surface unimodality statement. The geometric realisation is proved, not merely the matrix calculation, and a fixed seed gives infinitely many such examples.

36. Cycles refute infinite log-concavity for chromatic coefficients

[log-concavity-and-unimodality/infinite-log-concavity-of-cycles/article.pdf](#)

from [chromatic_cycle_logconcavity.zip](#)

For the cycle C_n with absolute chromatic coefficients under the iterated operator $L(a)_k = a_k^2 - a_{k-1}a_{k+1}$: infinitely log-concave for $3 \leq n \leq 11$; first failure at depth 5 for $n = 12$, depth 4 for $13 \leq n \leq 16$, and depth 3 for $n \geq 17$. So C_{12} is the smallest cycle counterexample (no minimum over all graphs is claimed), with an explicit third-iterate witness polynomial valid for all $n \geq 3$.

37. Oscillating Log-Concavity in Geometric Generating Functions

[log-concavity-and-unimodality/oscillating-log-concavity-geometric-transform/article.pdf](#)

from [oscillating_log_concavity_research.zip](#)

Challenges 2 and 3 in Section 3 of Heim and Neuhauser, arXiv:2302.13327v1, for the *geometric* transform $1/(1 - uG(z))$. For divisor-sum weights, every real exponent and every $u > 0$ give infinitely many strict log-concavity and log-convexity indices with eventually bounded gaps; likewise for integer power weights with exponent ≥ 2 , the exponents 0 and 1 being explicitly exceptional. For square weights at $u = 1$, every four consecutive indices carry both signs (sharp), each with density $1/2$ in every arithmetic progression. A subdominant-pole theorem is the common mechanism.

38. A counterexample and a sharp spacing-three criterion for products of q -integers

[log-concavity-and-unimodality/q-integer-product-unimodality/report.pdf](#)

from [q_unimodality_attack.zip](#)

The necessity half of Conjecture 5.4 of Connelly, Ito, Martinez, Shevchenko and Yang (arXiv:2605.12822v1) is false: $(1 + q)^6(1 + q^3)$ is symmetric and weakly unimodal although no a_i is divisible by 3 and $b = 2$ exceeds the asserted bound, and the example lies inside the source's claimed range. The report also proves the proposed sufficient condition for every $r \geq 2$, a necessary degree bound in the absence of an r -divisible ordinary factor, and a complete classification of spacings two and three.

39. Sharp size thresholds for failures of log-concavity in independence systems

[log-concavity-and-unimodality/independence-system-thresholds/article.pdf](#)

from [sharp_log_concavity_thresholds.zip](#)

For each rank $r \geq 4$, determines the smallest ambient ground-set size admitting failure of ordinary, ordered and ultra log-concavity, and shows the three thresholds coincide across all independence systems, those satisfying Xie–Xu's augmentation bound, and those with additive augmentation parameter 3 (values $7/6/5$, $7/7/6$, $8/7/7$, and $r + 1$ throughout for $r \geq 7$). The initial target, Xie–Xu Conjecture 4.8 (arXiv:2408.09152v1), had already been disproved by Alex Chengyu Li in September 2026; the report states this and refines the target to the rankwise minimum-size question.

40. A sharp order-five threshold for higher-order Stirling subset log-concavity

[log-concavity-and-unimodality/higher-order-stirling-rows/article.pdf](#)

from [stirling_log_concavity.zip](#)

For $S_{n,k}^{(r)}$, the number of partitions of $n + (r - 1)k$ labels into k blocks of size $\geq r$, proves that all rows are log-concave if and only if $1 \leq r \leq 5$, with strict inequalities in the interior — the log-concavity assertion of Deb–Sokal Conjecture 1.4(c) (arXiv:2507.18959v1). Also proves the cycle cases $r = 3, 4$; the fifth-order *cycle* conjecture is explicitly left unresolved, and the included operator counterexample is flagged as not being a counterexample to it.

41. A sum-of-squares proof of Ulas's real-zero conjecture

[log-concavity-and-unimodality/stern-polynomial-positivity/stern_conjecture_note.pdf](#)

from stern_conjecture_research.zip

Conjecture 4.4 of Maciej Ulas, arXiv:1909.10844v1, for the Stern polynomials indexed by $a_n = (4^n - 1)/3$ and $h_n = 2(4^n - 1)(2 \cdot 4^n + 1)/3 + 1$: for every $n \geq 0$ and every real t , $W_n(t) = (A_{n+1}(t) - tA_n(t)/2)^2 + 3t^2 A_n(t)^2/4 > 0$. The adjacent-polynomial quadratic identity behind it was already in Ulas's paper and is re-derived, not claimed as new.

8 Graph theory

42. Mutual-visibility sets in Andrásfai graphs

[graph-theory/andrasfai-mutual-visibility/article.pdf](#)

from andrasfai_visibility_research.zip

Proves the generating function attributed to Andrew Howroyd in OEIS A391632, $\sum a(n)x^n = x(4 - 19x + 33x^2 - 40x^3 + 8x^4)/(1 - 10x + 29x^2 - 32x^3 + 8x^4)$, by an exact weight-preserving bijection between mutual-visibility sets of the n -Andrásfai graph and anchored closed walks of length $3n - 1$ in a four-state labelled graph. Adds the full bivariate generating function, an exact binomial formula for every cardinality, maximum-set classification, eventual polynomiality at each fixed deficit, exponential asymptotics, a central limit theorem, and linear-time recognition and exact uniform sampling.

43. A bicyclic 30-vertex graph with domination root -4

[graph-theory/domination-root-minus-four-order-30/domination_root_30.pdf](#)

from domination_root_30_research.zip

An explicit connected planar graph on 30 vertices and 31 edges, treewidth 2 and domination number 10, with $D(G, x) = x^{10}(x + 4)Q_{19}(x)$ — a negative answer to question 2 of Section 4 in Alikhani and Griswold, arXiv:2608.00109v1, which asks whether 33 is the minimum order. Their 33-vertex example had already disproved the original integer-root conjecture; no claim of global minimality at 30 is made. Three independent exact polynomial computations agree.

44. A marked-cycle proof of the trapezohedral domination conjecture

[graph-theory/trapezohedral-minimal-dominating-sets/article.pdf](#)

from trapezohedral_domination_research.zip

Proves the generating function labelled conjectural in OEIS A381190: for the n -trapezohedral graph, the number of dominating sets that are inclusion-minimal among all dominating sets and induce a connected subgraph is $a_n = 2n(2p_{n-2} + p_{n-3})$ with $p_m = p_{m-2} + p_{m-3}$ (Padovan). The proof classifies such sets by a cyclic arrangement of gaps of length 2 and 3 with a restricted marked edge. Adds the size distribution, strict log-concavity, an unbiased sampler, a Binet formula and a central limit theorem.

9 Enumerative combinatorics

45. A Twelve-Value Proof of the A181280 Formula

[enumerative-combinatorics/a181280-binary-matrix-formula/article.pdf](#)

from A181280_research_package.zip

The binary-matrix formula recorded as Conjecture 20 by Kauers and Koutschan (2023) and still marked conjectural in OEIS A181280: an explicit spectral recurrence bound and a finite certificate resting on twelve decisive counts, together with further consequences. `STATUS.md` records that general C-finiteness was already proved in Kauers's December 2025 lecture slides.

46. Esthetic Numbers, Folded Paths, and Algebraic Diagonals

[enumerative-combinatorics/esthetic-numbers-and-folded-paths/article.pdf](#)

from esthetic_walks_research.zip

The diagonal conjecture $T(q, q - 2) = A182555(q - 2)$ in OEIS A377000, where $T(q, k)$ counts base- q positive integers of length k whose adjacent digits differ by one. Proves it exactly, proves

the entry's other two conjectures as retrieved, and derives formulas for every fixed-offset diagonal $q = k + d$.

47. Diagonal Hardinian Arrays: A Fixed-Parameter Asymptotic Formula

[enumerative-combinatorics/hardinian-array-diagonal-asymptotics/article.pdf](#)

from [hardinian_arrays_research.zip](#)

For every fixed $r \geq 0$, $H_r(n, n) \sim C_r 4^{rn} n^{-r(r-1)/2}$ with an explicit C_r in Gamma values. The formula confirms the constants tabulated by Dougherty-Bliss and Kauers (2024) but refutes their proposed restriction to powers of 2, 3 and π : $C_7 = 2^{1852}/(3^{45}\pi^3)$. Adds an exact finite determinant at fixed array size, an exact enumeration algorithm, and a limiting negative-binomial law for total deficits.

48. An explicit formula for lucky parking spots

[enumerative-combinatorics/lucky-parking-spots/article.pdf](#)

from [lucky_parking_spots_research.zip](#)

Proves Conjecture 13 of Butler, Hadaway, Lenius, Martens and Moats (JIS 29 (2026), Article 26.1.1): a closed form for $C(n, j)$, the number of parking functions in which *spot* j is lucky. Adds an exact reflection law, all fixed right-edge columns, a closed form for OEIS A374533, Poisson edge limits, uniform bulk convergence, polynomial reciprocity and asymptotic expansions.

49. Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations

[enumerative-combinatorics/adjacency-bounded-132-avoiders/article.pdf](#)

from [strict_growth_132.zip](#)

For 132-avoiding permutations with every adjacent difference at most m and growth constant α_m : proves $\alpha_{m+1} > \alpha_m$ for all $m \geq 1$ — the remaining strict-inequality part of Nadler's Conjecture 2 — and the refinement $4 - \alpha_m = (2 \log m + 4 \log \log m + O(1))/m$, with $2 \log \pi - 4 \log 2 \leq \liminf E_m \leq \limsup E_m \leq 2 \log \pi$. The all- m component ordering and convergence of E_m are explicitly not claimed.

50. Throwback Dynamics with Repeated Entries

[enumerative-combinatorics/throwback-queue-dynamics/throwback_research.pdf](#)

from [throwback_research.zip](#)

For the queue process that removes the head and reinserts it at the zero-based position equal to its weight: every token recurs if and only if at most m input tokens have weight at most m , for every $m \geq 1$, with an explicit description of the recurrent tokens when that fails. Adds exact cycle periods and frequencies, a coprimality criterion for perfect mixing, Catalan and prime-parking counts for recurrent cores, and the largest labelled period of a core of size r .

51. Reciprocity and Matrix Duality for Preorder Polytopes

[enumerative-combinatorics/preorder-polytope-reciprocity/article.pdf](#)

from [preorder_reciprocity.zip](#)

Proves Conjectures 7.2, 4.2 and 4.4 of Athanasiadis and Chapoton, *Polytopes and posets associated to preorders* (arXiv:2605.26916v1), and the ordinary $q = 1$ case of their Conjecture 4.9 (not its full q -refinement). Two mechanisms: Ehrhart–Macdonald reciprocity after subtracting the all-ones vector, extending the double identity to independently weighted capacities; and duplication of both colour classes of the bipartite graph followed by Postnikov's lattice-point formula, giving the conjectured matrix transpose.

52. Rationality and growth of stabilized permutation-weight series

[enumerative-combinatorics/stabilized-permutation-weights/article.pdf](#)

from [stabilized_permutation_weights.zip](#)

Studies the full stabilized generating functions of the permutation-weight q -Eulerian polynomials, beyond their initial connection with OEIS A256193: an all- d finite-spectrum theorem, rationality, a universal integer denominator, the exact radius of convergence and residue-class asymptotics. The two-descent sequence gets an explicit formula in powers of 3, powers of 2 and Fibonacci numbers.

10 Generating functions and asymptotics

10.1 Coefficient counts of binary products

Three reports on counting coefficients of $\prod_i(1+x^{G_i})$. The first two answer, independently, the universal-rationality part of Richard Stanley’s MathOverflow question 431075 (“A conjectured rational generating function”, 2022), which had no posted answer when checked on 19 September 2026. A third answer used the same mixed-base construction and the same enumerative argument as the first, and has been merged into it.

53. A Mixed-Base Counterexample to Stanley’s Rationality Question

[generating-functions-and-asymptotics/binary-product-coefficient-counts/mixed-base-counterexample/article.pdf](#)

from stanley_mixed_base_counterexample.zip + Stanley_Rationality_Counterexample.zip

With $G_{2j+1} = 2^j$ and $G_{2j+2} = 3^j$ the exponent generating function is rational, while $\nu(2n) = 2^{n-\lfloor(n+1)\alpha\rfloor}$ for $\alpha = \log_3 2$ counts the coefficients equal to one in $\prod_{i \leq N}(1+x^{G_i})$ and its generating function is not rational, not D-finite, not P-recursive, and has a natural boundary. Classifies rationality in the family interleaving a^j and q^j with digit polynomial $1+y+\dots+y^{a-1}$ for all $2 \leq a < q$ — Stanley’s general $r = a-1$ — as holding exactly when $q^u = a^v$. Adds a finite-quotient rigidity theorem, an abstract natural-boundary theorem for $\sum q^{\lfloor n\theta \rfloor} t^n$ at every irrational θ , the exact limit set of a_n/λ^n , and a streaming recurrence making the point that non-P-recursive does not mean hard to compute. *Merged; the enumerative half was the same argument in both originals, but the two routes to non-rationality differ and both are kept — an elementary finite-state reduction mod p , and the rigidity route that reaches the stronger transcendence conclusions.*

54. Rational exponents, nonrational coefficient counts

[generating-functions-and-asymptotics/binary-product-coefficient-counts/recursive-exponent-construction/research.pdf](#)

from stanley_counterexample.zip

A different negative answer, from the recursion $a_0 = 2$, $a_1 = 1$, $a_n = a_{n-1} - 2a_{n-2}$, with $b_n = T_{n-1} + 2^n - 1 + a_n$, $T_n = T_{n-1} + b_n$, $G_{2n-1} = 2^{n-1}$ and $G_{2n} = b_n$: the counting series has a natural boundary on the unit circle, hence is neither rational, algebraic nor D-finite.

55. Odd coefficients in quasifibonacci products

[generating-functions-and-asymptotics/binary-product-coefficient-counts/quasifibonacci-odd-coefficient-counts/article.pdf](#)

from quasifibonacci_parity_research.zip

Stanley’s Conjecture 6.3, with a seed-independent extension. For $k \geq 2$ and seeds each exceeding the sum of their predecessors, continued by the k -term recurrence, the number $h_k(n)$ of odd coefficients of $\prod_{i \leq n}(1+x^{w_i})$ has generating function $(1+2z^k)/(1-2z+2z^k-2z^{k+1})$, independently of the seeds. The proof builds periodic signs of block length $k+1$ and block product -1 and shows every signed finite product is flat.

10.2 Asymptotics, records and discrepancy

56. The Exponential Scale of Gregory Sign Thresholds

[generating-functions-and-asymptotics/gregory-coefficient-sign-thresholds/gregory_sign_threshold.pdf](#)

from gregory_sign_threshold_research.zip

For $(x/\log(1+x))^m = \sum_n G_n^{(m)} x^n$ and $\text{Gr}(m)$ the least N beyond which $(-1)^{n-1} G_n^{(m)} > 0$, proves $\log \text{Gr}(m) = m + 1 - \gamma - (\zeta(2) + \zeta(3))/m + C_2/m^2 + O(m^{-3})$ with C_2 explicit, plus a finite algorithm for every coefficient of the complete expansion; in particular $\text{Gr}(m)/e^m \rightarrow e^{1-\gamma}$ and $\text{Gr}(m)/3^m \rightarrow 0$. This replaces Conjecture 6.1 of Xu and Zhao (arXiv:2609.11072v1), whose printed lower bound already conflicts with its own reported $\text{Gr}(5) = 113$; that value is independently certified here.

57. Exact Records in the Square-Root-of-Three Walk

[generating-functions-and-asymptotics/sqrt-three-walk-records/article.pdf](#)
from `sqrt3_record_recurrence.zip`

Proves the four-phase record recurrence for $S(N) = \sum_{n \leq N} (-1)^{\lfloor n\sqrt{3} \rfloor}$ in the corrected form on page 7 of Bruin and Fokkink, *On the records and zeros of a deterministic random walk* (arXiv:2503.11734v2), where it is experimental and attributed to Arias de Reyna and van de Lune. Includes minimality of every claimed first passage, a rational generating function, closed forms, sharp logarithmic extrema, and an extension to every even integer trace $D \geq 4$.

58. Sharp discrepancy for a family of binary substitutions

[generating-functions-and-asymptotics/binary-substitution-discrepancy/article.pdf](#)
from `oeis_discrepancy_counterexamples.zip`

For the fixed word of $0 \mapsto 1, 1 \mapsto 101010$, the position sequences A284365 and A284366 both carry a conjectured upper position error below 2, and both conjectures are false: the first counterexamples are at ranks 2,977,771 and 8,933,313, with common error $(2977771\sqrt{21} - 13645857)/2 \approx 2.00487$ and sharp common upper bound $(6 + \sqrt{21})/5 \approx 2.11652$. Proves complete discrepancy limit intervals for every substitution $0 \mapsto 1, 1 \mapsto (10)^m$, with infinite counterexample families and exact recurrences for their indices.